Abdelrahman Khalifa

Embedded Software Engineer — Zephyr · Embedded Linux · C · Rust

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

SUMMARY

Embedded software engineer and the sole software engineer at EKSON, where I wrote and own the Zephyr RTOS firmware for pneumatic motion simulators. Embedded Linux experience with Yocto and Buildroot, a merged Zephyr networking contribution, and Google Summer of Code 2026 with The Linux Foundation. Writes C, C++ and Rust.

EXPERIENCE

Embedded Software Engineer — EKSON | **Aug 2025 – Present**

Sole software engineer for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for more deterministic timing, devicetree-based drivers and easier porting to other boards.
- Implemented closed-loop PID control of the pneumatic valves, with feedback from an MPU6050 IMU over I2C (calibration and a Mahony filter for orientation), plus safety stops and a watchdog.
- Built the PyQt5 desktop app that controls the simulator over a UART protocol; cut its CPU usage and latency by **~40%** (measured) by moving serial I/O off the UI thread.
- Built remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST, then flashes new firmware with esptool; GitHub Actions CI.

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | **May 2026 – Oct 2026**

COSMIC desktop printer setup tool in Rust | github.com/Abd002/GSOC-2026

- Built a Linux printer-management stack as 5 Rust crates with an async client, a CUPS/IPP/DNS-SD backend behind a varlink IPC service, and a shared UI; tested on real HP printers.
- Wrote Rust-to-C FFI bindings (libcups3 DNS-SD) and fixed crashes in libcups' Avahi code; **7 PRs merged upstream**.

Embedded Linux Systems Intern — Orange Digital Center (ODC) | **Dec 2023 – Jan 2024**

- Built Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing.

OPEN SOURCE

- **Zephyr RTOS** (Mar 2025): implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules (150+ lines), reviewed and merged upstream (issue #86977).
- **OpenPrinting libcups and cups-rs** (2026): fixed DNS-SD crashes in libcups (C); ported cups-rs to libcups3 with CI (PRs #18–#22, merged).
- **Rust ecosystem** (Dec 2025): Boa JavaScript engine string optimization (PR #4553) and a Rust Clippy lint fix (PR #16214), both merged.

PROJECTS

- **AUTOSAR-Based V2X Vehicle** (graduation project, STM32): I built the firmware update path (an ESP32/RPi gateway sends the image over UART) and a dual-bank bootloader with CRC check and rollback. GitHub
- **Seat Heater Control System** (Tiva C, FreeRTOS): real-time controller with temperature monitoring, heater-level control and UART diagnostics, using tasks, mutexes and event flags. GitHub
- **IoT Device Communication System** (C++, Raspberry Pi 4, QEMU, Yocto): TCP unicast and UDP multicast framework between a Pi server and emulated clients. GitHub
- **AUTOSAR PORT and GPIO drivers** (C, TM4C): MCAL drivers written and validated to the AUTOSAR spec.
- **Traffic Light Controller** (Verilog, Spartan-3E FPGA): FSM with debounce and delay modules. GitHub
- **Door Locker Security System** (C, two MCUs): password lock with keypad, LCD, I2C EEPROM and DC motor; UART between the MCUs. GitHub

TECHNICAL SKILLS

Languages: C, C++ (modern), Rust, Python, Bash, Verilog

RTOS & firmware: Zephyr (devicetree, west), FreeRTOS, ESP-IDF, OSEK/VDX (Trampoline), ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR, AUTOSAR, bootloaders, FOTA, PID control

Embedded Linux: Linux, Yocto, Buildroot, kernel modules, device drivers, device trees, QEMU, Raspberry Pi

Protocols & tools: UART, I2C, SPI, CAN, TCP/UDP, REST; Git, GitHub Actions CI, CMake, Make, PyQt5, debugging

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | **2020 – 2025**

Very Good (84.6%) with Honors. Codeforces Expert; 4-time ECPC finalist (2021–2024); IEEE student branch Treasurer.